Jihwan Bae

Ph.D. Student in Industrial Engineering (Human Factors)
Edwardson School of Industrial Engineering, Purdue University
Healthcare Ergonomics Analytics Laboratory (HEAL)
West Lafayette, IN, USA | bae129@purdue.edu

Research Interests

Human Factors; Healthcare Human–Robot Interaction; Physiological Sensing; Behavioral Modeling; Healthcare Decision-Making; AI-Assisted Human Performance Assessment

Education

Purdue University, West Lafayette, IN, USA — Aug. 2026 – Present
Ph.D. in Industrial Engineering (Human Factors), Expected 2030
Edwardson School of Industrial Engineering
Healthcare Ergonomics Analytics Laboratory (HEAL)
Advisor: Prof. Denny Yu

Korea Advanced Institute of Science and Technology (KAIST) — Sep. 2024 – Aug. 2026
Daejeon, South Korea
M.S. in Data Science

Korea University, Seoul, South Korea — Mar. 2019 – Aug. 2024
B.S. in Industrial & Management Engineering
B.E. in Artificial Intelligence

Research Experience

Graduate Research Assistant — Aug. 2026 – Present
Healthcare Ergonomics Analytics Laboratory (HEAL)
Edwardson School of Industrial Engineering, Purdue University

- Conduct research in human factors and healthcare applications under the supervision of Prof. Denny Yu.
- Investigate human performance and behavior using physiological, behavioral, and multimodal data.
- Current research focuses include physiological sensing, healthcare human–robot interaction, and AI-assisted assessment of human performance.

Visiting Researcher — Jun. 2025 – Aug. 2025
Healthcare Ergonomics Analytics Laboratory (HEAL), Purdue University

- Analyzed multimodal driving data, including EEG, ECG, and subjective measures, from a Purdue–Ford collaboration investigating driver alertness.
- Evaluated the effects of multisensory stimulation on driver vigilance using physiological and subjective measures.

Researcher — Apr. 2025 – Aug. 2026
National Medical Center (NMC) Research Project, South Korea

- Conducted statistical modeling of demand for essential healthcare services and infectious disease care under hypothetical pandemic scenarios.
- Contributed to quantitative modeling for healthcare demand and pandemic preparedness.

Research Intern — Dec. 2022 – Feb. 2023
Data Science & Artificial Intelligence Laboratory (DSAIL), KAIST

- Reviewed research on representation learning, recommendation systems, and graph-based machine learning.
- Contributed to research activities related to conceptual representation and inference in artificial intelligence.

Manuscripts Under Review

Bae, J., Nasir, M., Pitts, B., and Yu, D. *Within-Participant Associations of Heart Rate Variability and EEG With Self-Reported Alertness and Mood During In-Vehicle Multisensory Stimulation. Applied Ergonomics*, under review.

Bae, J., Kim, J., and Lee, T. *Modeling Experience Effects from Related but Non-Identical Choice Contexts. Journal of Choice Modelling*, under review.

Conference Proceedings

Kim, M., Kim, J., **Bae, J.**, and Lee, T. (2026). *Semi-Mechanistic Epidemic Scenario Projection with Empirical Temporal Dynamics.* In *Proceedings of the 2026 Winter Simulation Conference.*

Conference Presentations

Bae, J., Kim, J., and Lee, T. *Constructed Experience Measure and Hospital Choice: Decomposing Preference Heterogeneity and State Dependence.* 2026 KIIE Annual Spring Conference, Gyeongju, South Korea, June 2026.

Bae, J., Kim, J., and Lee, T. *Disentangling Experience and Attribute Effects in a State-Dependent Hospital Choice Model.* 2025 KIIE Annual Fall Conference, Daejeon, South Korea, November 7, 2025.

Kim, M., Kim, J., **Bae, J.**, and Lee, T. *Emulating Pandemic-X Scenarios: A Semi-Mechanistic Framework with Bayesian Filtering.* 2025 KIIE Annual Fall Conference, Daejeon, South Korea, November 7, 2025.

Honors & Awards

Burrer Family Award, Purdue University 2026 – 2030
Four-year scholarship awarded by the Edwardson School of Industrial Engineering in recognition of outstanding scholarly potential.

Veritas Research Scholarship, Korea University 2023
Selected in recognition of outstanding research achievement.

Semester High Honors, Korea University 2020, 2022, 2023

Veritas Educational Scholarship, NAVER Connect Foundation 2019
Selected for an iOS development education program.

Certificates

Oracle Certified Professional, Java SE 6 Programmer

Advanced Data Analytics Semi-Professional (ADSP)

Technical Skills

Programming and Data Analysis: Python, R, MATLAB, SQL

Additional Programming: C, Java, Swift

Scientific Computing and Documentation: LaTeX

Languages

Korean: Native **English:** Fluent